Unemployment and its Effects on Time Spent Engaging in the Community

By Jessica Buxbaum, Tim Hua, and Sam Segal*

Draft: May 27, 2021

This paper investigates the relationship between one's employment status and time spent on communal activities using 2006-2019 data from the American Time Use Survey (ATUS) and multiple linear regression. We define communal activities as participation in volunteer, civic, and religious organizations. These activities are important due to their positive associations with socially beneficial outcomes such as increased social trust and political participation—measures that are often broadly defined as social capital. Our findings suggest that those who participate in communal activities spend more time on them when they are unemployed. The effect is significant for communal activities in aggregate and volunteering activities for all individuals, and religious activities for women. However, becoming unemployed does not motivate people who otherwise do not participate in communal activities to participate in them, and it simply causes those who already spend time engaging with their community to spend more time doing so.

There are numerous benefits to engaging with one's community. Volunteering, for example, not only provides labor for charitable causes but also benefits the volunteer in numerous ways. The potential benefits include increased physical and mental health including longevity, reduced risky behavior in youths, and better occupational outcomes for women (Wilson and Musick 1999). Meeting and working with others in one's community builds trust and social capital, which in turn are associated with better functioning institutions and bureaucracies, as well as economic growth and financial development (Costa and Kahn 2003). The civic benefits of social engagement on

* Buxbaum: Middlebury, jbuxbaum@middlebury.edu. Hua: Middlebury, thua@middlebury.edu. Segal: Middlebury, ssegal@middlebury.edu

civic society has been documented in writings ranging from Alexis de Tocqueville's *Democracy in America* in 1835 to Robert Putnam's 1995 essay "Bowling Alone".

However, societal trust has fallen along with participation in communal activities (Uslaner 2000; Painter II and Paxton 2014). The national volunteer rate hit a fifteen-year low in 2015 after declining steadily since the early 2000s (Dietz and Grimm 2018). This decline is not entirely reflected in the amount of time the average American spends engaging in community activities, as seen in figure A1 and A2 (Hofferth et al. 2020).

This paper aims to investigate whether time spent on community-engaged activities is affected by whether one is employed or unemployed using the American Time Use Survey compiled by IPUMS. Unemployment adversely affects someone's emotional well-being and social connectedness. For example, unemployed individuals are sadder when engaging in leisure activities compared to employed individuals (Krueger and Mueller 2012). Communal activities, such as organized religion, can help moderate these difficulties (Bradshaw and Ellison 2010). We will also examine if unemployment impacts the proportion of individuals who volunteer, and use an interaction term to see if the effects of unemployment differ among men and women. This will help future researchers and policymakers who want to confront this societal trust crisis in designing interventions that involve unemployed persons.

It is difficult to pinpoint causality when studying communal activities and other outcomes. Recent literature generally rejects the notion that increased community engagement causes increased social capital outright due to inextricable simultaneous causality between the two, however the correlation between these two measures remains a significant area of interest and study. (Uslaner 2000).

This study is novel in several ways. The first is that we use ATUS data from 2006 up to 2019, the most recent year. The dataset contained information on how much time individuals spent on all sorts of communal activities, including volunteering, religious activities, and civic activities. This data takes place over a longer period compared to previous studies. The second is that we are able to measure both the rates at which individuals participate in communal activities and the time they spent volunteering. One previous study that examined volunteering behavior and employment is Piatak (2016) which focused only on volunteering rates.. Piatak used the 2003-2013 CPS data and found that unemployed individuals are significantly more likely to volunteer after adding controls.

This paper, however focuses on communal activities in general as an outcome variable, which is also rather novel compared to those solely studying volunteering or religion. Previous studies on time use of the unemployed usually looked at categories that tend to occupy more of someone's time such as leisure and non-market work–all categories where the unemployed spend more time compared to employed individuals (Krueger and Mueller 2012; Aguiar, Hurst, and Karabarbounis 2013). Aguiar and colleagues, for example, examined how individuals substitute between homemade goods produced non-market work and goods bought on the market.

I. Data Description

We used data from the American Time Use Survey (ATUS) made available through IPUMS.org. ATUS is conducted by the U.S. Census Bureau for the Bureau of Labor Statistics (BLS) as an add-on to the monthly Current Population Survey. Households that completed the CPS that represent a range of demographic characteristics are selected, and one member of the household aged 15 or over is asked to respond to the ATUS on a voluntary basis. Respondents are asked to provide a time log of their activities in a 24-hour period over the phone. Time spent on each activity is recorded in minutes.

We selected data from 2006 to 2019 and examined time use variables classified under the bls_social tag, which includes time spent on volunteering, civic engagement, and religious activities—activities we consider to be communal. We used the standard employment variable from the CPS, which classifies individuals as those who are employed and at work, employed and absent from work, unemployed because of a layoff, unemployed and actively looking, and not in the labor force. We created a dummy variable for employment by categorizing all employed people (either at work or absent) and all unemployed people (on layoff or looking for work). Although individuals not in the labor force make up a significant portion of the population, they are not included in our research in order to focus on the impact of unemployment on time spent on communal activities specifically.

We also examined the three major subcategories under bls_social: time spent volunteering (bls_social_volact), time spent on religious activities (bls_social_religious), and time spent on civic activities (bls_social_civic).

In total, we have data on 104,024 individuals, 97,514 employed and 6,510 unemployed. We did

not want to overfit the model and used data on race, gender, age, family income, and whether or not someone lives in a metro area as control variables. We recoded race into four dummy variables for individuals who are Hispanic, White and non-Hispanic, Black and non-Hispanic, and individuals who are any other race and non-Hispanic. We dropped those who are under 18 from our analysis, and the age in the sample ranges from 18 to 85. CPS codes everyone from age 80-84 as 80, and everyone at or above the age of 85 as 85. Given that less than one percent of the participants fell into this category, we felt no need to adjust for this. Family income is coded into 16 brackets. Details of the income brackets, as well as additional information on the variables used, can be found in table A4 in the appendix, while summary statistics can be found in table A1.

II. Model

To assess the effect employment has on how long an individual spends on an activity and how the effect could be different for male and females, we use the following linear model with an interaction term.

$$\text{(1)} \quad \text{Time Spent on an Activity}_i = \beta_0 + \beta_1 * \text{Employment}_i + \beta_2 * \text{Female}_i + \beta_3 * \text{Employment}_i * \text{Female}_i + \beta_4 * \text{Family Income}_i + \beta_5 * \text{Age}_i + \beta_6 * \text{Metro}_i \\ \textit{Race Dummy Variables} + \textit{Time Fixed Effects} + \textit{State Fixed Effects} + u_i$$

We also wanted to examine the mechanism through which employment affects how much time people spend on these activities. Specifically, when compared to their employed counterparts, are unemployed individuals more/less likely to engage in an activity, or are they more/less likely to spend more time in that activity? To answer this question, we examine the proportion of individuals who spent any time on a given day in a communal activity using a linear probability model.

$$\text{(2)} \quad \text{Spent Time on Activity}_i = \beta_0 + \beta_1 * \text{Employment}_i + \beta_2 * \text{Female}_i + \beta_3 * \text{Employment}_i * \text{Female}_i + \beta_4 * \text{Family Income}_i + \beta_5 * \text{Age}_i + \beta_6 * \text{Metro}_i \; \textit{Race Dummy Variables} + \textit{Time Fixed Effects} + \textit{State Fixed Effects} + u_i$$

We chose linear models because the results are easy to interpret and because it is the standard used in other papers (Aguiar, Hurst, and Karabarbounis 2013; Krueger and Mueller 2012). If the reader is curious, all results cease to be statistically significant when the dependent variable is transformed using $ln(\textit{Time Use Variable} + 1)$. We did not use logit/probit models for similar reasons.

III. Results

Table A2 reports the results from our main regression. We elected to not report the results with no fixed effects because the differences between the two are not significant. During our time period of study, unemployment is associated with spending more time engaging with one's community. This effect is significant for social activities as a whole and volunteer activities. It is also significant for women engaging in religious activities. In other words, being unemployed increases the amount of time women spend on religious activities significantly, but the effect on men is not significant. None of the coefficients on civic activities are significant. This is partially explained by the fact that participation in many civic activities, like voting and jury duty, are mandatory.

We found other notable results of this regression, including how women tend to spend more time in social and religious activities. There are also significant racial differences. Non-White individuals spend more time on religious activities. F-tests revealed that Black persons spend more time engaging with their communities in general compared to the other three racial groups ($p = 0.000$). Hispanic individuals and those who are neither Hispanic, White, nor Black tend to spend less time volunteering.

Finally, note that family income is not significantly correlated with how individuals spend their time and those in metropolitan areas spend less time engaging in their community.

The results of the linear probability regression is in table A3. Employment seemed to have no significant impact on how often different individuals engage in communal activities. The two regressions lead us to the following conclusion: Being unemployed is associated with an increase in the amount of time spent in communal activities in aggregate and volunteering activities for both males and females, as well as an increased time spent on religious activities for females. This increase is caused not by more people engaging in these activities but more time spent from those who already engage in these activities.

A. Commentary on results

Results of the regression may be viewed as insignificant given that the values seem quite small—one may consider a four minute decrease in an activity as not realistically significant. However, given how the ATUS is administered, it is inevitable that the vast majority of answers for smaller categories will be zero. These numbers represent a daily average. Another way to think about our results is the following: There are currently about 160 million individuals in the U.S. labor force, of which 6.1% are unemployed (U.S. Bureau of Labor Statistics 2021a, 2021b). If those who are unemployed spent the same amount of time engaging in various communal activities as their employed counterparts, we would lose about 732,000 hours of communal engagement and 395,000 hours of volunteering *every day* in the U.S. That is certainly significant. Similar interpretation methods could be applied to the differences between ethnic groups, and future researchers can look into racial differences in time use.

Unemployed individuals spend more time on volunteering and communal activities, giving encouraging signs that unemployed individuals seek opportunities for increased social capital. However, future policymakers who aim to increase social capital must not forget that workplace interactions, which are unavailable to unemployed persons, are a vital source of social capital that could very well overshadow the effects from communal activities investigated here.

Finally, we would like to acknowledge that differences between the results surrounding volunteer rates and results surrounding volunteer time are a great example of how time use data can uncover trends that we have not previously revealed.

B. Potential Omitted Variable Bias

While we controlled for individuals' gender, race, income level, age, and whether or not someone lives in a metro area, other omitted variables may create bias in our coefficients. If an individual has an illness or health condition, they may be less likely to find a job that can suit their needs and and spend time on these types of activities, causing a positive bias on the coefficient. Additionally, we did not control for the type of industry that an individual works in. Some industries may have less jobs available, and some may allow their employers more time to spend on community-engaged activities. Additionally, some professions work with more people, which may cause individuals not to participate in outside community-engaged activities. We also cannot control for whether or not the individuals' parents participated in community-engaged activities. If individuals saw their parents participate, or even spent time with their parents in these activities at a young age, they will be more likely to participate. These variables may cause positive or negative biases on our coefficients. However, we are generally confident in our findings since many of the omitted variables are expected to be correlated with household income (for example, sick individuals may earn less money), which we found to not significantly affect our findings. In fact, removing income from the regression barely affected our coefficients and standard errors (e.g. the coefficient on employment for time spent volunteering changed from -2.430 to -2.424, and the standard errors stayed the same at 1.157).

IV. Conclusion

We sought to identify patterns in the way employed and unemployed people participate in community-engaged activities, or more specifically volunteering, civic obligations, and religious activities. By conducting a multiple regression with an interaction term and controlling for age, gender, household income, race, whether an individual lives in a metro area, as well as state and year fixed effects, we were able to find several statistically significant results. While becoming unemployed does not change whether or not an individual participates in these types of activities, it does change the amount of time they spend. There are clear benefits associated with participation in community-engaged activities, and policy should focus on increasing participation rates in these types of activities for unemployed and employed people alike.

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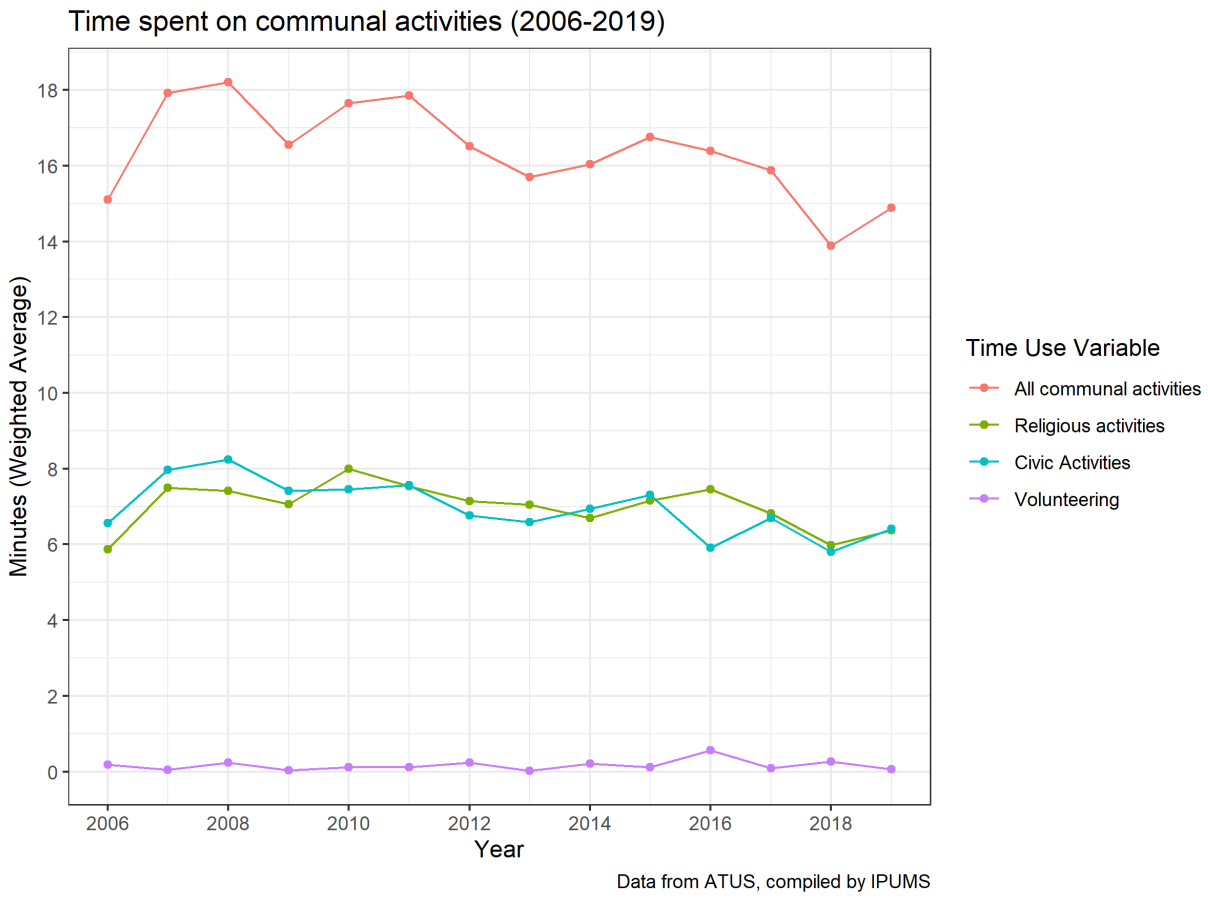
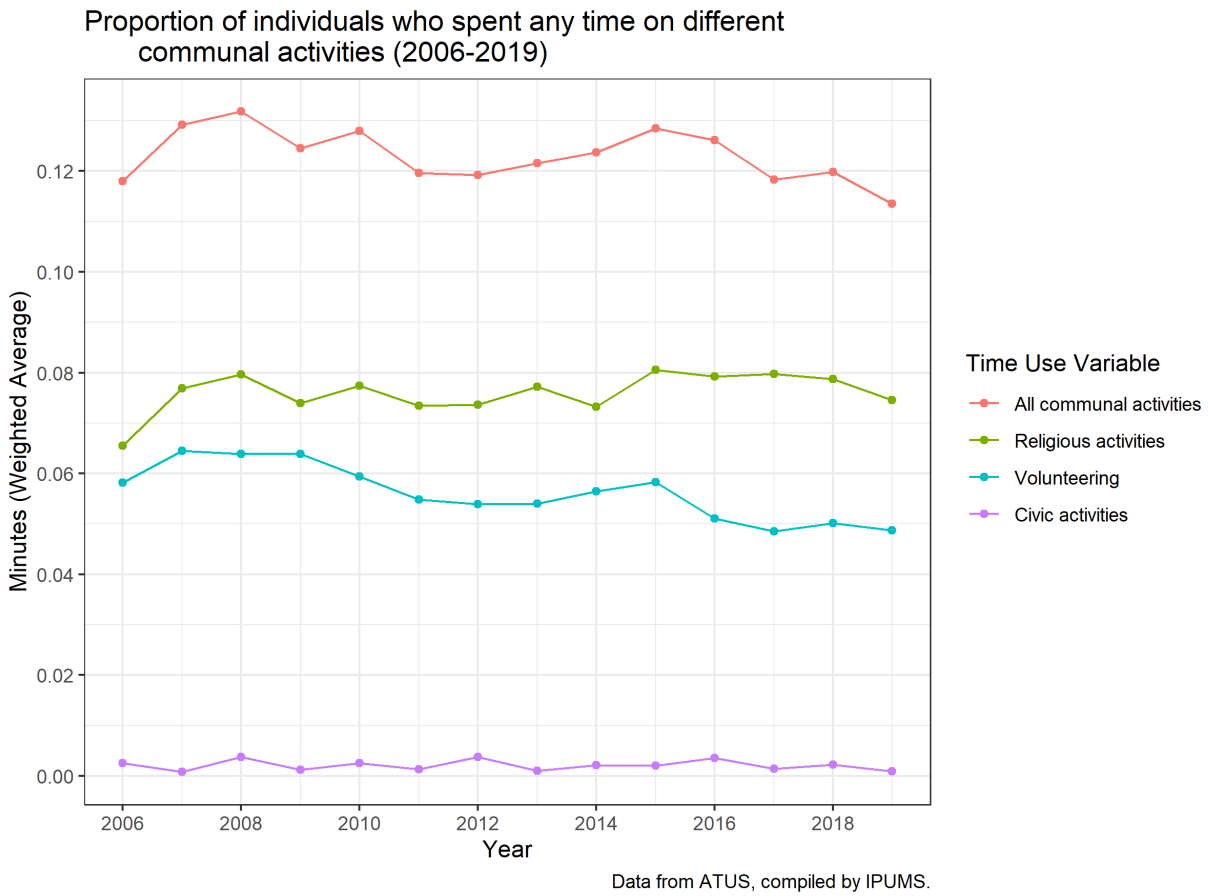
Tables and figures

Figure A1.



Source: (Hofferth et al. 2020).

FIGURE A2.



Source: (Hofferth et al. 2020).

Table A1—Summary means across Employed and Unemployed Persons

Variables	Means for Employed (n=97,514)	Means for Unemployed (n=6,510)
Total time spent on communal activities	14.83	15.81
Proportion who spent any time on communal activities	0.12	0.10
Time spent on civic activities	0.14	0.00
Proportion who spent any time on civic activities	0.001582	0.00
Time spent on religious activities	6.27	8.59
Proportion who spent any time on religious activities	0.08	0.08
Time spent on volunteering activities	6.37	5.10
Proportion who spent any time volunteering	0.05	0.04
Family income category (more information is in the appendix)	12.42 ($50,000 - $74,999)	10.70 ($35,000 - $49,999)
Proportion who are female	0.47	0.50
Age	42.47	36.13
Proportion who are White and non-Hispanic	0.65	0.52
Proportion who are Black and non-Hispanic	0.11	0.15
Proportion who are of another Race and non-Hispanic	0.07	0.06
Proportion who are Hispanic	0.17	0.26
Proportion who live in a metro area	0.87	0.88

Source: American Time Use Survey compiled by IPUMS (Hofferth et al., 2020)

Table A2—Results from Multiple Regression on Time Spent on Community-Engaged Activities

	1	2	3	4
Variables	All Communal	Religious	Volunteering	Civic
Employment	-4.502**	-0.450	-2.430**	-1.270
	(1.823)	(0.614)	(1.157)	(1.029)
Female	5.735**	3.756***	2.834*	-1.332
	(2.470)	(0.950)	(1.708)	(1.026)
Employment *Female	-2.607	-2.200**	-1.428	1.320
	(2.509)	(0.977)	(1.733)	(1.029)
Household income	-0.00122	8.51e-05	-0.00112	6.32e-05
	(0.00107)	(0.000557)	(0.000742)	(0.000108)
Age	0.324***	0.146***	0.137***	-0.00147
	(0.0172)	(0.00838)	(0.0119)	(0.00263)
Black Non-Hispanic	8.088***	7.164***	-0.183	0.118
	(0.816)	(0.474)	(0.501)	(0.115)
Other Non-Hispanic	0.514	2.226***	-1.689**	-0.144
	(0.957)	(0.460)	(0.658)	(0.138)
Hispanic	0.306	3.367***	-3.108***	-0.104
	(0.687)	(0.387)	(0.417)	(0.168)
Metro	-2.380***	-1.353***	-0.737	-0.247
	(0.753)	(0.339)	(0.476)	(0.318)
State and time fixed effects	Yes	Yes	Yes	Yes
Constant	9.026***	3.130**	3.346*	1.518
	(2.916)	(1.384)	(1.779)	(1.316)
Observations	104,024	104,024	104,024	104,024
R-squared	0.011	0.013	0.006	0.003
Robust standard errors in parentheses				
*** p<0.01, ** p<0.05, * p<0.1				

Table A3—Results from Multiple Regression on Proportion of Individuals Who Spent Any Time on Community-Engaged Activities

	(1)	(2)	(3)	(4)
VARIABLES	All Communal	Civic	Religious	Volunteering
Employment	-0.00622	-0.000636	-9.62e-05	-0.00528*
	(0.00714)	(0.00477)	(0.00523)	(0.00278)
Female	0.0387***	0.0186***	0.0319***	-0.00595**
	(0.00993)	(0.00693)	(0.00754)	(0.00285)
Employment* Female	-0.00493	-0.00516	-0.00870	0.00659**
	(0.0103)	(0.00717)	(0.00782)	(0.00288)
Household income	-2.36e-06	-6.37e-06	1.83e-06	1.72e-07
	(6.51e-06)	(4.71e-06)	(4.99e-06)	(1.11e-06)
Age	0.00245***	0.00104***	0.00168***	3.06e-06
	(9.64e-05)	(6.88e-05)	(7.55e-05)	(1.54e-05)
Black Non-Hispanic	0.0458***	-0.0120***	0.0608***	0.00119
	(0.00440)	(0.00272)	(0.00378)	(0.000794)
Other Non-Hispanic	-0.000214	-0.0224***	0.0232***	-0.000995
	(0.00515)	(0.00335)	(0.00428)	(0.000782)
Hispanic	-0.00257	-0.0309***	0.0258***	-0.000154
	(0.00385)	(0.00244)	(0.00323)	(0.000605)
Metro	-0.0147***	-0.00284	-0.0137***	-0.000505
	(0.00388)	(0.00269)	(0.00303)	(0.000840)
State and time fixed effects	Yes	Yes	Yes	Yes
Constant	0.0324**	0.0175*	0.00855	0.00577
	(0.0142)	(0.00969)	(0.0110)	(0.00355)
Observations	104,024	104,024	104,024	104,024
R-squared	0.019	0.011	0.019	0.002
Robust standard errors in parentheses				
*** p<0.01, ** p<0.05, * p<0.1				

Table A4—Variable Descriptions

Variable	Description
Household Income	Family income buckets: 1 Less than $5,000 2 $5,000 to $7,499 3 $7,500 to $9,999 4 $10,000 to $12,499 5 $12,500 to $14,999 6 $15,000 to $19,999 7 $20,000 to $24,999 8 $25,000 to $29,999 9 $30,000 to $34,999 10 $35,000 to $39,999 11 $40,000 to $49,999 12 $50,000 to $59,999 13 $60,000 to $74,999 14 $75,000 to $99,999 15 $100,000 to $149,999 16 $150,000 and over 996 (Refused), 997 (Don't know), and 998 (Blank) were all coded as missing
All Communal Activities	"Time spent actively participating in religious and spiritual activities, volunteer activities, and civic obligations and participation as well as waiting and travel associated with these activities"
Civic	"Time spent on civic activities, including court responsibilities, participating in government surveys and voting, communicating with political representatives, police officers, and judges, applying for relief and social services, and getting licenses"
Religious	"Time spent on religious activities, including attending religious ceremonies, services, and life events, attending retreats and celebrating holidays, participating in religious group meetings, praying, and educational activities."
Volunteering	"Time spent on volunteering activities, including computer use, organizing, collecting, and preparing materials, conducting telephone calls, publicizing, fundraising, preparing food, leading organizations and teaching."

Source: American Time Use Survey compiled by IPUMS (Hofferth et al., 2020)